

Dijkstra's algorithm

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1 **Initialization:**

2 $N' = \{u\}$

3 for all nodes v

4 if v adjacent to u

5 then $D(v) = c(u, v)$

6 else $D(v) = \infty$

7

8 **Loop**

9 find w not in N' such that $D(w)$ is a minimum

10 add w to N'

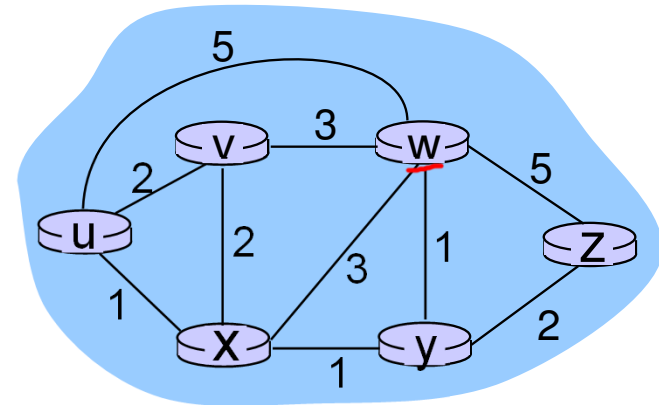
11 update $D(v)$ for all v adjacent to w and not in N' :

12 $D(v) = \min(D(v), D(w) + c(w, v))$

13 /* new cost to v is either old cost to v or known

14 shortest path cost to w plus cost from w to v */

15 **until all nodes in N'**



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9 find n^* not in N' such that $D(n^*)$ is a minimum

10 add n^* to N'

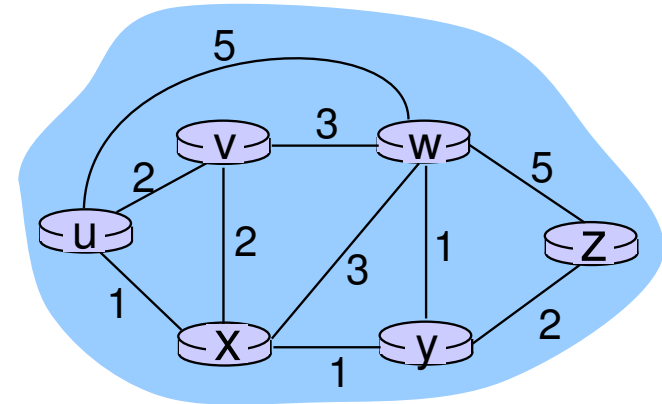
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15 **until all nodes in N'**



Step	N'	D(^v v),p(v)	D(^w w),p(w)	D(^x x),p(x)	D(^y y),p(y)	D(^z z),p(z)
0						
1						
2						
3						
4						
5						

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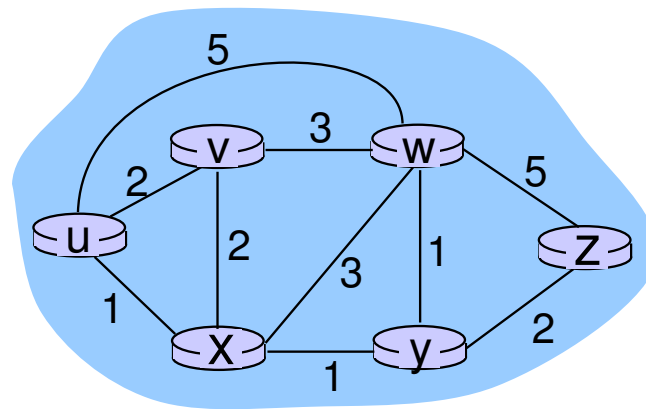
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Step	N'	D(v),p(v)	D(w),p(w)	D(x),p(x)	D(y),p(y)	D(z),p(z)
0	u	2, <u>u</u>	5, <u>u</u>	1, <u>u</u>	∞	∞
1						
2						
3						
4						
5						

Source = u

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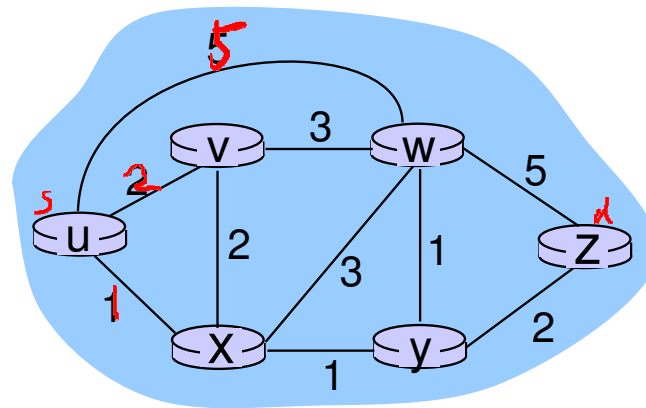
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$n^* = x$ $D(n^*) = 1$

Step	N'	$D(v), p(v)$	$D(w), p(w)$	$D(x), p(x)$	$D(y), p(y)$	$D(z), p(z)$
0	u	2, u	5, u	1, u	∞	∞
1	u, x	$\min\{2, 1+2\} = 2$, u	$\min\{5, 1+3\} = 4$, x	1, u	$\min\{1+1, \infty\} = 2$, x	∞
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3						
4						
5						

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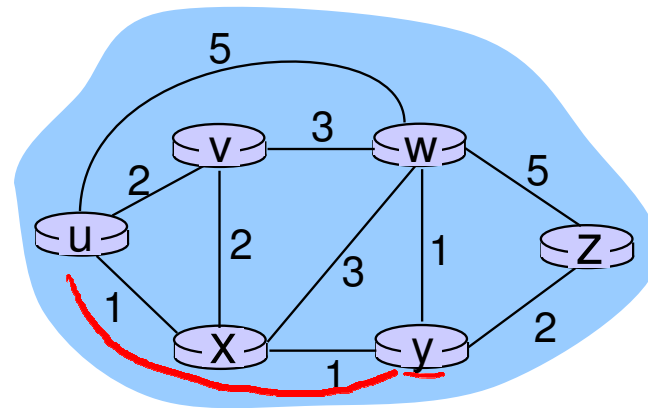
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$n^*=y$

Step	N'	$D(v),p(v)$	$D(w),p(w)$	$D(x),p(x)$	$D(y),p(y)$	$D(z),p(z)$
0	u	2, u	5, u	1, u	∞	∞
1	u,x	$\min\{2, 1+2\}$ =2, u	$\min\{5, 1+3\}$ =4, x		$\min\{1, 1+1\}$ =1, y	∞
2	u,x,y					
3						
4						
5						

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